

ICL GEN AI MASTERY

Mini Assessment- 1

Customer Support Ticket Analysis Workflow

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Course: GEN AI Mastery

1. Overview

This assessment demonstrates a three-step Generative AI workflow for analyzing a customer support complaint. The workflow progressively converts an unstructured customer message into structured, actionable information. Step 1 extracts important complaint details, Step 2 determines urgency and priority, and Step 3 combines the results into a valid JSON record.

Workflow at a Glance

Input	Customer complaint in natural language
Step 1	Extract key customer-support details
Step 2	Assess urgency and priority
Step 3	Create a structured JSON output
Final Result	A concise, machine-readable support ticket summary

2. Step 1 - Information Extraction

Purpose: Identify the most important facts from the customer's complaint without adding information that was not provided.

Prompt :

I want you to act as a customer support assistant.

Read the customer complaint given below and pick out the important details from it.

I mainly want: Customer name , Main problem, Product or service involved,Any money-related issue, How the problem is affecting the customer, Any deadline mentioned

Here is an example:

Example 1:

Customer complaint:

"Hi, I am Priya. I cannot log in to my office email even after changing my password."

Important details:

Customer name: Priya

Main problem: Cannot log in

Service: Office email

Money-related issue: None

Effect on customer: Cannot access email

Deadline: Not mentioned

Now read the following complaint and give me the important details in the same simple format.

Customer Complaint Input

"My name is Dude. I made a payment of Rs. 11,000 yesterday, but the money was taken from my bank account and my order still shows as unpaid. I need this fixed before tomorrow because I need the order ASAP."

Step 1 Output

Important details:

Customer name: Dude

Main problem: Payment made, but order still shows as unpaid

Product/Service: Order

Money-related issue: Rs. 11,000 was deducted from the bank account

Effect on customer: Cannot get the order processed and needs it urgently

Deadline: Before tomorrow

3. Step 2 – Urgency and Priority

Purpose: Classify the complaint using the defined Low, Medium, and High urgency levels and assign an operational priority.

Prompt:

I want you to look at the details found from a customer complaint and decide how urgent the problem is.

Use only these three levels:

Low - The problem is small and can wait.

Medium - The problem is important but does not need immediate action.

High - The problem needs quick attention because of things like money loss, account problems, major service issues, or a close deadline.

First, look at the details given below.

Then give:

- Urgency: Low, Medium, or High

- Reason: A short and simple reason
- Priority: Immediate, Normal, or Low

Here are the details found in the first step:

Important details:

Customer name: Dude

Main problem: Payment made, but order still shows as unpaid

Product/Service: Order

Money-related issue: Rs. 11,000 was deducted from the bank account

Effect on customer: Cannot get the order processed and needs it urgently

Deadline: Before tomorrow

Now decide the urgency based on those details.

The details extracted in Step 1 were supplied to the second prompt.

Step 2 Output

- **Urgency:** High
- **Reason:** ₹11,000 has already been deducted, the order is still unpaid, and the customer needs it processed before tomorrow.
- **Priority:** Immediate

Step 3 – Structured JSON Generation

Purpose: Combine the customer and issue from Step 1 with the urgency and priority from Step 2, then produce a valid JSON object using exactly five required fields.

prompt:

I want you to put the information from the first two steps into one simple JSON format.

Use exactly these five fields:

customer
issue
urgency
priority
summary

Take the customer and issue from Step 1.

Take the urgency and priority from Step 2.

For the summary, write one short sentence explaining the whole problem.

Make sure:

- The JSON is valid.
- Do not add any extra text.
- Do not add an explanation.
- Do not use a code block.
- Do not make up any information.

Step 1 result:

Important details:

Customer name: Dude

Main problem: Payment made, but order still shows as unpaid

Product/Service: Order

Money-related issue: Rs. 11,000 was deducted from the bank account

Effect on customer: Cannot get the order processed and needs it urgently

Deadline: Before tomorrow

Step 2 result:

- **Urgency:** High
- **Reason:** ₹11,000 has already been deducted, the order is still unpaid, and the customer needs it processed before tomorrow.
- **Priority:** Immediate

Now give me only the final JSON.

Step 3 Output

```
{  
  "customer": "Dude",  
  "issue": "Payment made, but order still shows as unpaid",  
  "urgency": "High",  
  "priority": "Immediate",  
  "summary": "₹11,000 was deducted from the bank account, but the order is still unpaid and needs to  
be processed before tomorrow."  
}
```

4. Observations

- Generative AI can extract useful information from an unstructured customer complaint.
- Breaking a larger task into smaller prompts improves the clarity and usefulness of each stage.
- A separate classification step can convert extracted facts into an actionable urgency and priority decision.
- Structured JSON makes the final result easier for software systems, dashboards, or ticketing workflows to consume.
- Clear instructions and fixed output formats help reduce unnecessary text and keep the workflow consistent.

5. Conclusion

The Customer Support Ticket Analysis Workflow demonstrates how Generative AI can support customer-service operations through a simple, repeatable pipeline. Starting from a natural-language complaint, the workflow extracts relevant details, evaluates urgency, assigns priority, and produces a structured JSON result. This provides a practical example of using prompt design and multi-step reasoning to transform raw customer input into actionable support information.